#1

Solution. First review the fact about $l : P^*E \to \mathbb{C}$, defined as

$$l(P^*\phi) := \int_0^T (f(t,\cdot), P^*\phi(t,\cdot))_{s,-s}\, dt - i(g, \phi(0,\cdot))_{s,-s}$$

Well defineness and continuity has shown in the note. As P^*E is still a subspace of L^2H^{-s}, for $\overline{P^*E}$, we have $L^2H^{-s} = \overline{P^*E} \bigoplus \overline{P^*E}^{\perp}$. Then for $P^*\phi_j \in P^*E$ converge to $P^*\phi$ in its closure, we first define

$$l(P^*\phi) := \lim_j l(P^*\phi_j)$$

This limit is independent choice of sequence as

$$|l(P^*\phi_j) - l(P^*\phi_k)| \le C|P^*\phi_j - P^*\phi_k|_{L^2H^{-s}}$$

. Next, for $h \in L^2H^{-s}$, as $h = h_1 + h_2$ where $h_1 \in \overline{P^*E}, h_2 \in \overline{P^*E}^{\perp}$, then for $h \in L^2H^{-s}$, define the extension as

$$l(h) := l(h_1) \quad \text{where } h_1 \in \overline{P^*E}$$

well defineness preserves as above decomposition is unique. Now we have desired map. Last step is to figure out linearity and continuity. Then we can apply Riesz Repr., so everything keeps clear. Linearity is trivial, and continuity hold as $|l(h)| \le C|h_1|_{L^2H^{-s}} \le C|h|_{L^2H^{-s}}$. Last inequality hold by parallelogram law. ■

#2

Solution. Need to show exist $C > 0$, s.t. $|u|_{CH^s} \le C|u|_{H^1H^s}$. Here we have $u, \partial_t u \in L^2H^s$, we first find where to use sobolev embedding. Here we have $C^1H^s \subset H^1H^s$ and dense under topology of H^1H^s then we first shown such fact in C^1 rather than proof in H^1, which will occur some problem that is not easy to solve. (Note density is attained through density of $\mathcal{S}$)

So first we shown that $u \in C^1H^s$, we have

$$|u(t,\cdot)|_s^2 - |u(\tau,\cdot)|_s^2 = 2Re\int_\tau^t (u'(r,\cdot), u(r,\cdot))_s dr \le 2|\partial_t u|_{L^2H^s}|u|_{L^2H^s}$$

Take integral w.r.t. τ, we have

$$T|u(t,\cdot)|_s^2 \le \int_0^T |u(\tau,\cdot)|_s^2 d\tau + 2T|\partial_t u|_{L^2H^s}|u|_{L^2H^s}$$

As $2ab \le a^2 + b^2$, we have

$$|u(t,\cdot)|_s^2 \le (1/T + 1)|u|_{L^2H^s} + |\partial_t u|_{L^2H^s}$$

By abitrarily choose t, this imply C^1H^s works. Then apply density pick a sequence $\{u_j\} \subset C^1H^s$ converge to some $u \in H^1H^s$ then that cauchy sequence imply a uniform limit $v$in CH^s, this 2 limit are same a.e.. so we are done. ■